

WEEK 17 – Architect of a Zoo

Grade: Intermediate (9) **Unit:** Measurement and Geometry

Curriculum Expectations

MPM 1D/MFM 1P: solve problems involving the areas and perimeters of composite two-dimensional shapes

SEL: recognize that testing out different approaches to problems and learning from mistakes is an important part of the learning process, and is aided by a sense of optimism and hope

Activity

- 1) For this activity, you are the architect of a new zoo and you are to design the animal exhibits to meet the requirements needed for the animals and the requirements of the land that you own as the zoo.
- 2) The contractors that you are working with have pre-set designs that they will use to create the animal exhibits with pre-set areas. You may combine these shapes together in any way you would like to make composite shapes. The animals have minimum surface area requirement for their exhibit depending on the animal and the zoo has a maximum amount of land that can be used based on the property it sits on. These are listed on the activity sheet below.
- 3) Through a process of estimating, trial and error, you must use the set designs to develop a zoo that meets the requirements, assembling shapes together within the zoo's boundaries. You can assume that the walkway is run between the exhibits, so leave a little space for them.
- 4) Once you have completed the zoo design, test to see if you can make another or if there is a more efficient way to meet the requirements

Check for Understanding

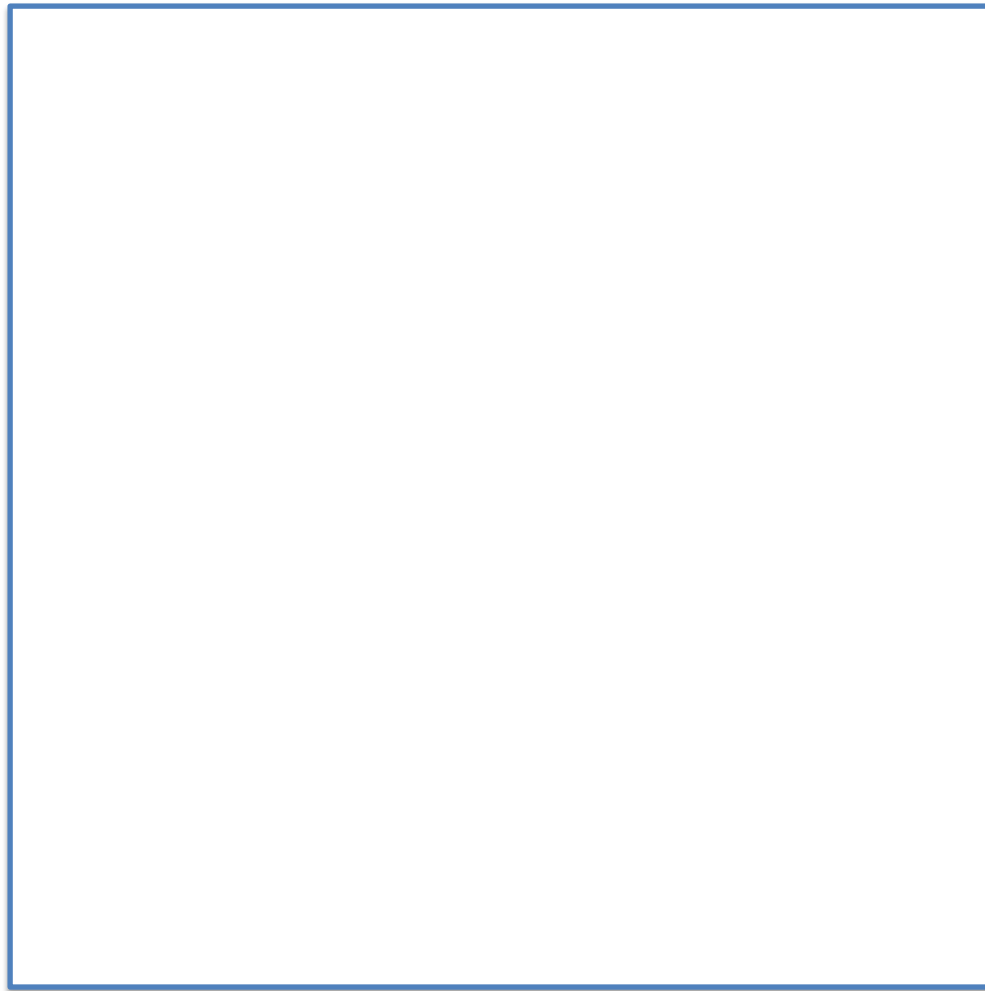
- I understand how to create composite shapes to meet set criteria
- I can calculate the area and perimeter of composite shapes
- I understand the relevance of calculating area and perimeter in real life situations

Materials

Recording sheet (attached below), pencil, information sheet below or internet access, calculator

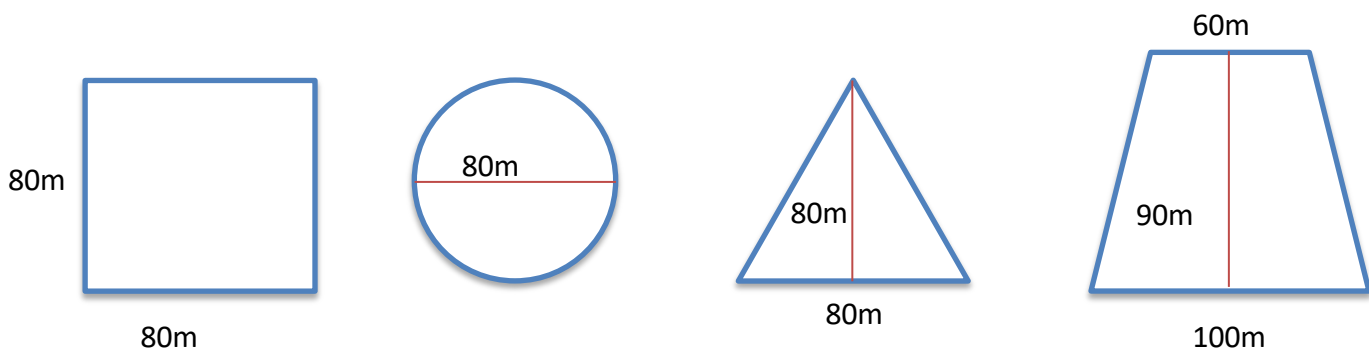
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The Zoo's boundaries measure in a square 500 metres x 500 metres. You may sketch your final configuration of animal exhibits in the above layout

Below are the contractors set designs for animal exhibits. You will need to combine these shapes to meet the needs of the various animals as outlined in the chart below.





Animal Exhibit Requirements

Animal	Minimum Surface Area Required	Special Requirements
Monkeys	12,500 m ²	
Tigers	20,000 m ²	
Giraffes	20,000 m ²	
Petting zoo animals	25,000 m ²	Must take the form of a rectangle for easy entrance and exit
Hippo	17,500 m ²	Must have a circular pool for the animals to swim in
Bears	17,500 m ²	Must have a circular pool for the animals to swim in
Pandas	12,500 m ²	

Recording Sheet

Animal	Number and Shapes Used to Meet Requirements	Sketch of Design	Area and Perimeter of the composite shape



$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\sum_{k=1}^n 2k = n(n+1)$$



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